

Indications, Safety, and Warnings

StealthStation™ S8 navigation platform

The StealthStation™ S8 system is intended as an aid for precisely locating anatomical structures in either open or percutaneous surgical procedures. The StealthStation™ system is indicated for any medical condition in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the skull, a long bone, or vertebra, can be identified relative to a CT or MR-based model, fluoroscopy images, or digitized landmarks of the anatomy.

The StealthStation™ S8 system with StealthStation™ cranial software is intended as an aid for locating anatomical structures in either open or percutaneous procedures. Their use is indicated for any medical condition in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as a skull, can be identified relative to images of the anatomy. This can include, but is not limited to, the following cranial procedures (including stereotactic frame-based and stereotactic frame alternatives-based procedures):

- Tumor resections
- Cranial biopsies
- General ventricular catheter placement
- Pediatric ventricular catheter placement
- Depth electrode, lead, and probe placement
- General localization

The StealthStation™ S8 system, with StealthStation™ S8 spine software, is intended as an aid for precisely locating anatomical structures in either open or percutaneous neurosurgical and orthopedic procedures. Their use is indicated for any medical condition in which the use of stereotactic surgery may be appropriate, and where reference to a rigid anatomical structure, such as the spine or pelvis, can be identified relative to images of the anatomy.

This can include the following spinal implant procedures, such as:

- Pedicle screw placement
- Iliosacral screw placement
- Interbody device placement

Residual risks

Residual risks related to the use of a Medtronic computer-assisted surgery system and its associated applications are the same as the residual risks for the procedures in which the system is used.

Contraindications

Medical conditions that contraindicate the use of a Medtronic computer-assisted surgery system and its associated applications include any medical conditions that may contraindicate the medical procedure itself.

Warnings

- The system hardware and software should be used only by qualified medical professionals who are trained in performing surgery and familiar with image-guided surgery systems.
- If system navigation seems inaccurate and recommended steps to restore accuracy are not successful, abort use of the system.
- Inspect all visible system components before use including the localizers that accompany your system (that is, the camera, the electromagnetic (EM) instrument interface, the side-mount emitter, or the flat emitter). If visibly damaged, do not use the system.
- To avoid the risk of electrical shock, do not simultaneously touch the patient and the system input/output panel, mouse, keyboard, or batteries. To avoid the risk of electric shock, this equipment must only be connected to a supply mains with protective earth.
- Metallic and conductive objects in or near the electromagnetic navigation field can degrade navigational accuracy and navigation volume. If metallic distortion causes excessive error, navigation may be disabled. To restore navigation, move metallic and conductive objects away from the navigation field.

- Electrical noise in or near the electromagnetic navigation field can degrade navigational accuracy. If electrical noise introduces excessive error, the system may automatically disable navigation. To restore navigation, move devices that produce electrical noise (such as electrocautery equipment and electric drills) away from the navigation field.
- Prevent fluid from entering any part of the StealthStation™ S8 system. Shut down the system, disconnect the power, and allow the system to dry if you suspect fluids may have entered any part of the system.
- Do not step on the base or legs of the system carts because the carts could tip over and cause injury.
- The system is not suitable for use in the presence of a flammable, anesthetic mixture with air, oxygen, or nitrous oxide.
- Do not modify the StealthStation™ S8 system without prior written consent from Medtronic Navigation, Inc. Prohibited modifications include but are not limited to altering, repairing, or replacing system components; altering software provided by Medtronic; updating the operating system or drivers; and installing any software that is not provided by Medtronic, including antivirus software. Secure the system to prevent unauthorized modifications. Unauthorized modifications could render the system unsafe or ineffective for its intended use.
- Follow system component maintenance and care instructions.
- The StealthStation™ S8 system camera cart contains a stereo system for music playback from a user provided audio player. This system may be capable of sound levels which can lead to permanent hearing loss at high volume. The volume is controlled at the user-provided audio player. For hearing safety, follow all instructions and warnings prescribed by the manufacturer of the user-provided player.

Stealth™ tractography

Because tractography is based on the diffusion of water in the brain, any condition that alters the flow of water may alter the fiber tract results. For example, the presence of edema around a tumor, which is like a pool of water, can result in an incomplete tract. Because fiber tracts cross one another throughout the brain, tractography may result in a tract that crosses from one fiber tract to another or an incomplete tract. This limitation is more prevalent when using the standard diffusion tensor imaging (DTI) tract fidelity.

Precautions

- Federal law (United States) restricts this device to sale by or on the order of a physician.
- Caution: The system and its associated applications contain no user-repairable parts. For repair or replacement of any part of the system or application, contact a Medtronic Navigation™ technical support representative.
- To avoid interrupting navigation, do not use the StealthStation™ S8 system on battery power alone while the system is in use with a patient.
- Periodically plug in the system to recharge the backup batteries. The backup batteries on the main cart and the camera cart are intended to keep the system operating for a short time if power is lost or disconnected. If the backup batteries are not charged, the system will shut down immediately when power is lost. The backup battery in the camera cart will recharge only when the two carts are connected to the cart-to-cart cable.
- Before surgery, clean the system carts and other non-sterilizable system components according to the parameters in the StealthStation™ S8 system cleaning instructions (9735727).
- The system mouse and keyboard are not designed for sterilization and may be damaged if sterilization is attempted.
- System components are fragile. Use care when handling system components.
- Keep the rear storage bin clear of extraneous materials which could block airflow through the perforated panel. Blocking air flow could result in excessive internal cart temperatures, which could degrade the system's performance and longevity.
- Avoid spilling fluids onto casters because fluid may impair the braking ability and the stability of the cart at rest on inclined surfaces.